1 PULL-UP RESISTORS ON DIGITAL INPUT

On the module RobDuino we can find two “on-board push button switches”. Wiring of this switches is presented in fig. 1, where can be noticed that both switches are connected to ground potential.

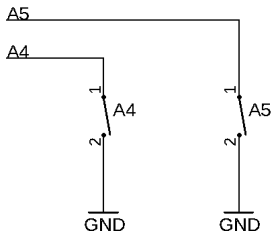

Figure 1: Wiring of on-board switches.

To properly use this push-button switches we must enable the `pull-up` resistors of A4 and A5 input of microcontroller.

1.1 Tasks:

1. Configure pins `A4` and `A5` as inputs with `pull-up` resistor.
2. At the end of the `setup()` function add the **`while`**`-loop` which will delay the execution of the program until we press the `A4` key.
3. Use the `A5` key for stop the robot and terminate the execution of the program.

Program 1: Pullup resistor on digital input.

```
void setIOpins();
void robotStop();
void robotForward();
void robotBackward();
void robotLeft();
void robotRight();

void setup(){
  setIOpins();
}

void loop() {
    robotForward();
    //to-do: the key reading
    bool stopTheRobotKey = 0;
    if (stopTheRobotKey == 1)
    {
        robotStop();
        exit(0);        //terminate the program
    }
}
void setIOpins(){
  pinMode(LEFT_MOTOR_PIN_A, OUTPUT);
  pinMode(LEFT_MOTOR_PIN_B, OUTPUT);
  pinMode(RIGHT_MOTOR_PIN_A, OUTPUT);
  pinMode(RIGHT_MOTOR_PIN_B, OUTPUT);
  pinMode(BUMPER_PIN, INPUT);
  //to-do: config the A4 and A5 pins

  //to-do: while-loop waiting for A4 is pressed

}
...
```

1.2 Questions:

1. What is the programming instruction of reading the value form digital input?
2. Which values can be assigned to **`bool`** type variable?
3. Explain the programming instruction `exit(0)`.

1.3 Summary:

1.3.1 <++>

<++>

1.4 Issues:

1.4.1 <++>

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